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ODD END SEMESTER EXAMINATION DECEMBER- 2025

Name of the Course: B.Tech. (ME)

Semester: **V**

Name of the Paper: Robotics

Paper Code: **TME 513**

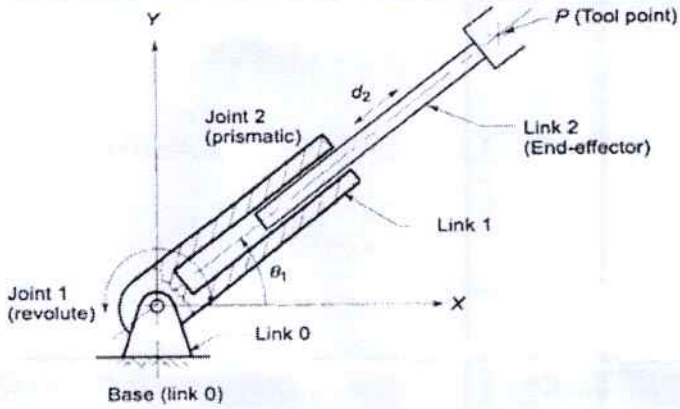
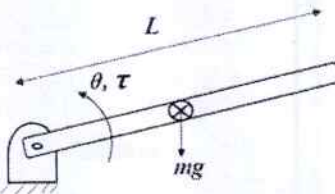
Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	CO1
(a)	Describe the following in brief: i) robot degrees of freedom and ii) laws of robotics.	
(b)	What are robotic actuators and sensors? Explain their types, working principles, and significance in robot motion and feedback control.	
(c)	Define a robot end effector. Discuss the types of end effectors used in robots, explaining the difference between grippers and tools, along with examples of mechanical, vacuum, magnetic, and adhesive grippers.	
Q2	(20 marks)	CO2
(a)	Define a coordinate frame in robotics. Explain the difference between global (world) and local (robot) coordinate frames with suitable diagrams.	
(b)	A frame {Q} is rotated relative to a frame {P} about z axis by 45 degrees, translated 3 units in x, and translated 5 units in y; the z axis is pointing out of the page. Position vector of a point with respect to frame {Q} is given by $[5 \ 3 \ 0]^T$. Calculate the position of that point with respect to frame {P}.	
(c)	A frame {B} is rotated relative to another frame {A} about Z axis by 60 degrees and translated 5 units in X and 6 units in Y. The Z axis is pointing out of the page. Thus, we have a description of ${}^A_B T$. Find ${}^B_A T$.	
Q3	(20 marks)	CO3
(a)	Discuss the steps involved in deriving the forward kinematic equations of a serial manipulator using the DH convention. Illustrate with a 2-link planar manipulator example.	
(b)	Formulate the inverse kinematic model for the 2-DoF, RP planar manipulator shown in the figure below.	

		
(c)	A 2-DOF planar manipulator has a revolute joint at the base (θ_1) followed by a prismatic joint (d_2) which extends along the direction of link-1. Link-1 has fixed length of 2.0 m. The desired end-effector position in the base frame is $P(x, y) = (2.5, 1.5)$ m. Find the joint variables θ_1 and d_2 that place the end-effector at P.	
Q4	(20 marks)	
(a)	What do you understand by the static force analysis of manipulators? Also, discuss about the role of Jacobian matrix in the static force analysis.	CO4
(b)	What do you understand by the manipulator differential motion? Also, describe the relationship between Jacobian matrix, joint velocities vector, and the end-effector velocities vector.	
(c)	For a 2-DOF planar RR manipulator, the link lengths are $l_1 = 1$ m and $l_2 = 1$ m. If $\theta_1 = 45^\circ$ and $\theta_2 = 30^\circ$, find the Jacobian matrix J relating joint velocities to end-effector linear velocity.	
Q5	(20 marks)	
(a)	Formulate the dynamic model of the one degree of freedom manipulator shown in the following figure using the Lagrange-Euler formulation. 	CO5
(b)	What are the main components of the dynamic model of a robot manipulator? Briefly describe the roles of: Inertia matrix Coriolis and centrifugal forces Gravity vector	
(c)	What do you understand by the manipulator dynamics? Discuss about the Lagrange-Euler and the Newton-Euler formulations for the manipulator dynamics.	